

EX.NO:4 CREATE AN ABSTRACT CLASS – SHAPE CLASS

PROGRAM:

```
import java.util.Scanner;
abstract class LibraryMember {
int memberId;
String name, email, phone;
LibraryMember(int memberId, String name, String email, String phone) {
this.memberId = memberId;
this.name = name;
this.email = email;
this.phone = phone;
}
void display() {
System.out.println("Member ID : " + memberId);
System.out.println("Name : " + name);
System.out.println("Email : " + email);
System.out.println("Phone : " + phone);
}
abstract void generateSummary();
}
class StudentMember extends LibraryMember {
StudentMember(int id, String name, String email, String phone) {
super(id, name, email, phone);
}
void generateSummary() {
System.out.println("\n----- Student Member -----");
display();
System.out.println("Borrowing Limit : 5 Books");
System.out.println("Penalty Per Day : Rs.2");
System.out.println("Annual Membership Fee: Rs.200");
}
}
class FacultyMember extends LibraryMember {
FacultyMember(int id, String name, String email, String phone) {
super(id, name, email, phone);
}
void generateSummary() {
System.out.println("\n----- Faculty Member -----");
display();
System.out.println("Borrowing Limit : 10 Books");
System.out.println("Penalty Per Day : Rs.3");
System.out.println("Annual Membership Fee: Rs.500");
}
}
class ExternalMember extends LibraryMember {
ExternalMember(int id, String name, String email, String phone) {
super(id, name, email, phone);
}
void generateSummary() {
System.out.println("\n----- External Member -----");
display();
System.out.println("Borrowing Limit : 3 Books");
System.out.println("Penalty Per Day : Rs.5");
System.out.println("Annual Membership Fee: Rs.1000");
}
}
```

```

public class Main {
public static void main(String[] args) {
Scanner sc = new Scanner(System.in);
System.out.print("Enter Member ID: ");
int id = sc.nextInt();
sc.nextLine();
System.out.print("Enter Name: ");
String name = sc.nextLine();
System.out.print("Enter Email: ");
String email = sc.nextLine();
System.out.print("Enter Phone: ");
String phone = sc.nextLine();
System.out.println("\nSelect Member Type");
System.out.println("1. Student");
System.out.println("2. Faculty");
System.out.println("3. External");
System.out.print("Enter Choice: ");
int choice = sc.nextInt();
switch (choice) {
case 1:
StudentMember s = new StudentMember(id, name, email, phone);
s.generateSummary();
break;
case 2:
FacultyMember f = new FacultyMember(id, name, email, phone);
f.generateSummary();
break;
case 3:
ExternalMember e = new ExternalMember(id, name, email, phone);
e.generateSummary();
break;
default:
System.out.println("Invalid Choice");
}
sc.close();
}
}

```

OUTPUT:

```

Enter Member ID: 101
Enter Name: Janaki
Enter Email: janaki@gmail.com
Enter Phone: 9876543210
Select Member Type
1. Student
2. Faculty
3. External
Enter Choice: 1
----- Student Member -----
Member ID : 101
Name : Janaki
Email : janaki@gmail.com
Phone : 9876543210
Borrowing Limit : 5 Books
Penalty Per Day : Rs.2
Annual Membership Fee: Rs.200

```